

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

Prof. Francesco Bripi

Derivatives: Forwards and Futures (Chapter 24)



Context: Why Derivatives exist

- ✓ Derivatives are a financial instruments designed to manage risk (hedging) ... we will see how later
- ✓ Here is Warren Buffett's definition:
 - *“Essentially, these instruments call for money to change hands at some **future date**, with the amount to be determined by one or more reference items, such as interest rates, stock prices or currency values”*
- ✓ So: financial contracts **whose price and cash flows** are determined by other, **underlying items...** and **delayed delivery!**

Context: derivatives and the Financial Crisis

- ✓ You may have heard that derivatives played an important role in the Financial Crisis of 2008
- ✓ Indeed, Warren Buffett, in his **2002** (!) letter to shareholders said*:
 - *“Charlie [Munger, Berkshire’s vice chairman] and I, view them as **time bombs**, both for the **parties** that deal in them and the **economic system**.”*
 - *... “In our view, however, **derivatives are financial weapons of mass destruction**, carrying dangers that, while now latent, are potentially lethal.”*
- ✓ He said this 5-6 years before the crisis happened!

Context: why Derivatives exist

- ✓ So, do derivatives *contain* or *amplify* risk?
- ✓ Both views are true:
 - derivatives hedge risk for the market as a whole ...
 - ... but can also concentrate risk in the hands of a few (*counterparty risk*)
- ✓ If these few (ex: AIG) are interconnected (linked) to everyone a small problem (subprime mortgage default) can lead to a collapse of the system!

Chapter Preview

✓ There are many derivatives. We will be looking at:

1. Forwards (today)

2. Futures (today and next class)

3. Options (next class and the one after the next)

✓ Many other derivatives exist:

- interest-rate swaps, CDS, CDOs...

Let's look at the **hedging** property first

Hedging

Hedging:

- ✓ is the process of sterilizing your risks by entering additional financial contracts which have **risk** that is **opposite** to your original risk
- ✓ involves engaging in a financial transaction that reduces or eliminates risk

Idea is to enter another financial contract with a risk that is opposite (negatively correlated) with the original risk

- ✓ Useful to define two notions for the “sides” of risk
 1. **Long** position
 2. **Short** position
- ✓ We literally say “you’re **long/short** in asset X”

Hedging

- ✓ If you have a **long** position (say: “*you are long*”)
 - think you buy an asset in the future but want to fix the price today
 - you **profit** if the asset’s price increases in the future (*buy cheaper*)
 - you **run** the risk that the asset’s price goes down
 - ex: same if you already own the asset

- ✓ If you have a **short** position (say: “*you are short*”)
 - think you sell an asset (you already have[@]) but want to fix its price today
 - you **profit** if the asset’s price decreases (*sell at higher price*)
 - you **risk** that the asset’s price goes up
 - ex: if the item is a production input for your business
 - @: **short selling**: you don’t own the asset yet today, so you borrow it to make the sale and return the asset to original owner in the future

Hedging

- ✓ Recall: *hedging* means to enter another financial contract with a risk opposite to the original risk
- ✓ You **hedge**
 - ... a *long* position by entering a *short* position
 - ... a *short* position by entering a *long* position
 - recall hedging is taking a risk that is opposite to original risk
- ✓ When you hedge a *long* position by entering a *short* position
 - the risk that the asset's price goes *down* is balanced by opposite risk in short position (price goes *up*)
- ✓ when you hedge a *short* position by entering a *long* position
 - ... vice versa

Example of Hedging (non-financial)

- ✓ It is Sep 2021. Suppose you own an oil company
- ✓ You have just been given the rights to extract oil from an oil field in Texas from September 2022
- ✓ Your cost of production is \$25/bbl (read bbl as “per barrel”)
- ✓ The current market price (spot price, S_t) is around \$80
- ✓ You’re a smart entrepreneur and want to hedge the risk that the market price will fall below \$25
 - ... it was around \$20 just one year ago!
- ✓ They’d be happy to enter a **short** position to hedge

OIL PRICE (Brent Crude)



Example of Hedging (non-financial)

- ✓ A utility company want to purchase oil to produce electricity
- ✓ They are happy with the **spot price** of \$80, but worry that geopolitical tensions may increase it
- ✓ Say: “they are short in oil”
- ✓ They’d be happy to enter a **long position to hedge**
- ✓ We could write a contract between the oil company and the utility company ...
 - that fixes prices of oil today (Sep 21) for delivery and payment at future date (Sep 22)

Example of Hedging (non-financial)

- ✓ Contract between the Oil company and the Utility company
 - the Oil company promises the delivery of 10,000 barrels for \$80/bbl (delivery price) at future date (Sep 2022)
 - the Oil company would still make a profit of \$55/bbl ($=\$80 - \25)
 - the Utility company locks in today's price at \$80

Example of Hedging (non-financial)

- ✓ Contract between the Oil company and the Utility company
 - the Oil company promises the delivery of 10,000 barrels for \$80/bbl (delivery price) at future date (Sep 2022)
 - the Oil company would still make a profit of \$55/bbl ($=\$80 - \25)
 - the Utility company locks in today's price at \$80

F (price agreed at $t = 0$ for delivery in T): $= \$80$



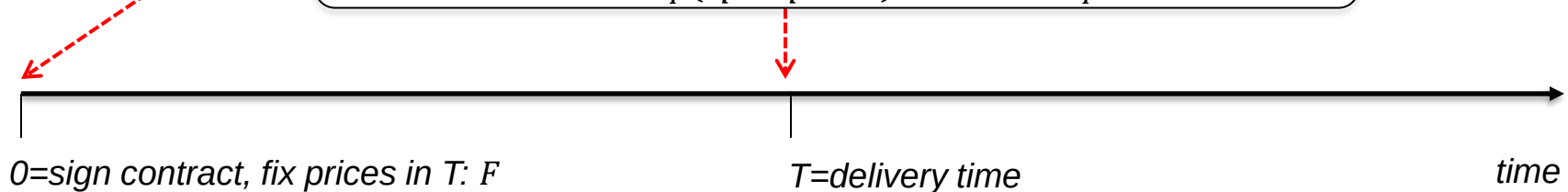
Example of Hedging (non-financial)

- ✓ Contract between the Oil company and the Utility company
 - the Oil company promises the delivery of 10,000 barrels for \$80/bbl (delivery price) at future date (Sep 2022)
 - the Oil company would still make a profit of \$55/bbl ($=\$80 - \25)
 - the Utility company locks in today's price at \$80

F (price agreed at $t = 0$ for delivery in T): $= \$80$

Case 1: S_T (spot price) $= \$100$: $S_T > F$

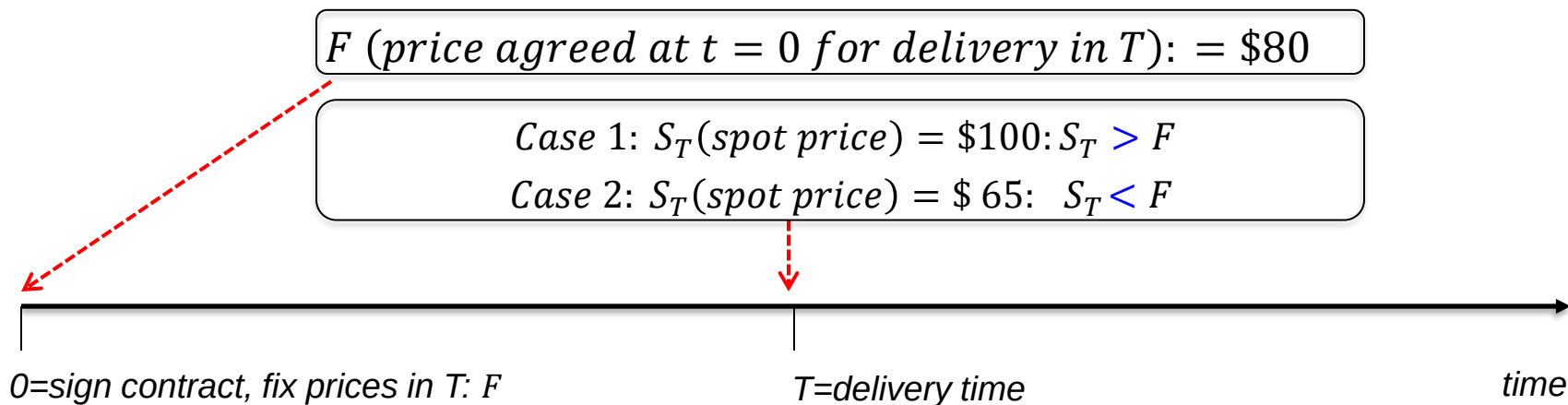
Case 2: S_T (spot price) $= \$65$: $S_T < F$



Q: Who wins in each case?

Example of Hedging (non-financial)

- ✓ Contract between the Oil company and the Utility company
 - the Oil company promises the delivery of 10,000 barrels for \$80/bbl (delivery price) at future date (Sep 2022)
 - the Oil company would still make a profit of \$55/bbl ($=\$80-\25)
 - the Utility company locks in today's price at \$80



Q: Who wins in each case?

A: – Case 1: Utility company, it buys at a price lower than market price: $S_T > F$
– Case 2: Oil company, it sells at price higher than market price: $S_T < F$
– reverse for losers: Oil in case 1, Utility in case 2

Forward Contract

1. If the spot price of oil in September 2022 is \$100 ($>$ \$80):
 - the *utility* company has profited from the forward because it locked in a lower price (\$80): it **profits \$20** ...
 - ... but the *oil* company **lost \$20** because the spot price \$100 is above forward price (\$80)
 2. Reverse is true if spot price in Sep 2022 is \$65 ($<$ \$80)
 - the *utility* company has **lost \$15** from the forward because the spot price is below the forward price (\$80)
 - the *oil* company **profited \$15** because the spot price \$65 is below forward price (\$80)
- ✓ **Key points** about **derivatives** in general:
- *ex-ante* they reduce risk for both players ...
 - ... but it is a **zero-sum game** (for every winner there's a loser)
 - each party sacrifices some profits in one state of the world so each can *benefit from insurance*

Forward Contract

- ✓ This is an example of a **forward contract**
- ✓ An agreement to enter a transaction at some future date
- ✓ It is not standardized in any way (it is “customized”), but at least it should specify:
 1. **What** item should be delivered and **quality** (properties of the item)
 2. **Quantity** to be delivered
 3. **Price** (delivery price)
 4. **Date** of delivery
- ✓ Eliminates uncertainty about the future **price ex-ante**, but...
- ✓ ... **ex-post**: there is always one winner and one loser!

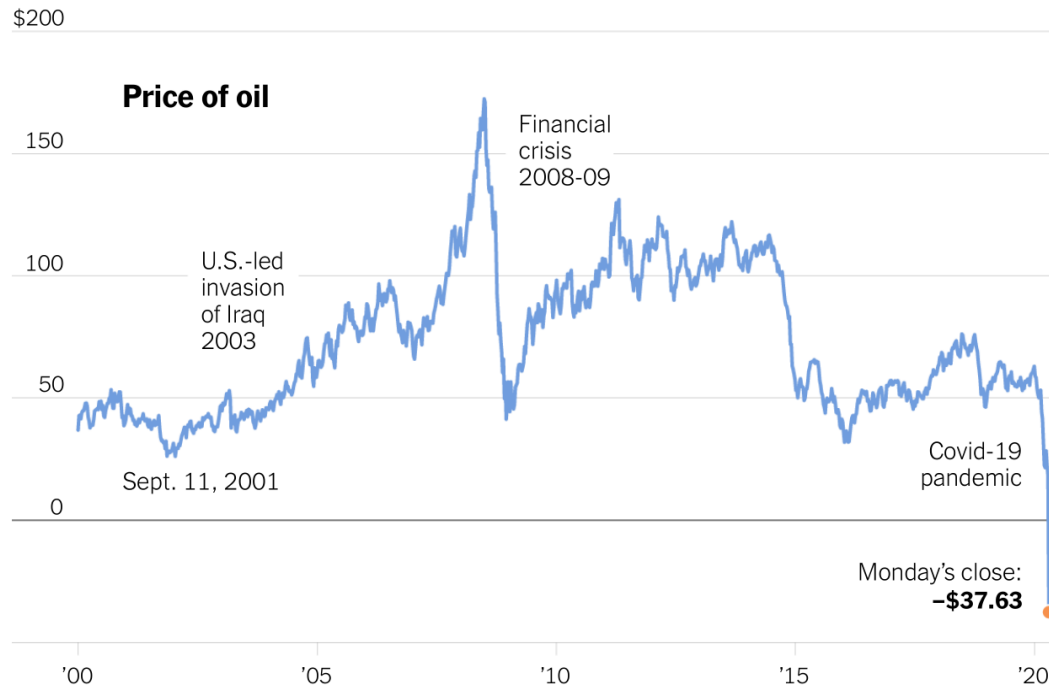
Who won the bet?



Oil prices and Covid-19

Too Much Oil: How a Barrel Came to Be Worth Less Than Nothing

One oil price went negative on Monday, signaling that there is no place to store all the crude the world is producing but not using.

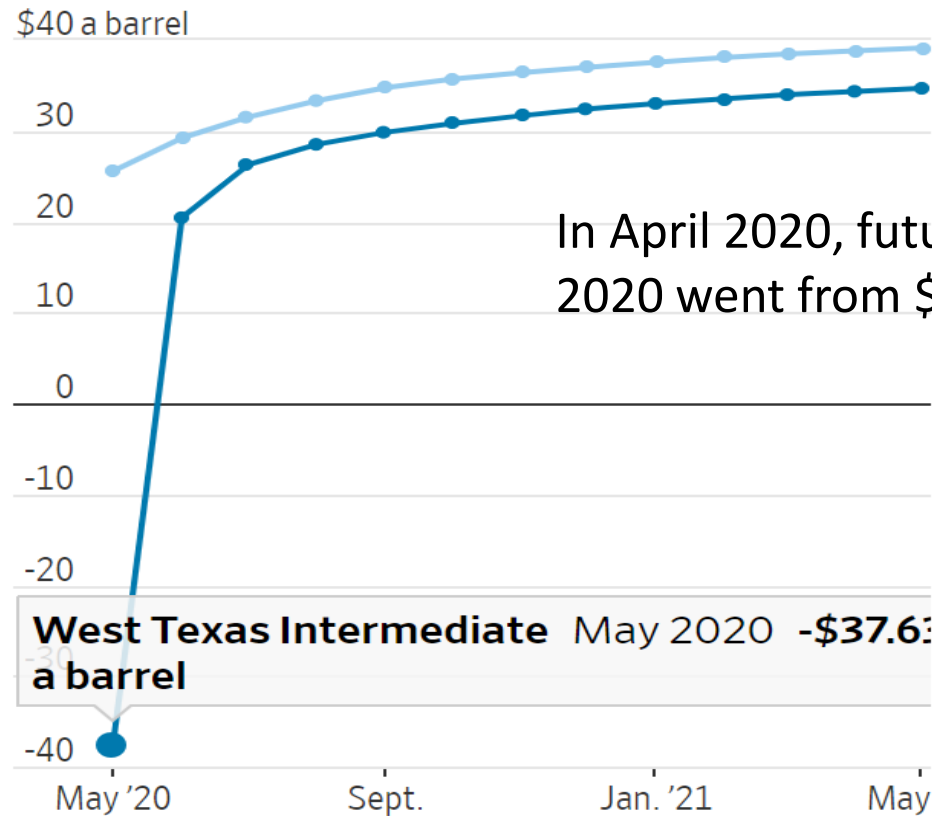


WTI crude, adjusted for inflation, plotted weekly • Source: Refinitiv • By The New York Times

Future prices for May 2020 delivery (in April 2020)

Oil futures price by month delivered

■ West Texas Intermediate ■ Brent



In April 2020, future price for May 2020 went from \$18 to -\$37!

West Texas Intermediate May 2020 **-\$37.63**
a barrel

Profitability of a Forward Contract

- ✓ Profits of the party **selling** a forward contract:
 - in our example, this was the Oil company
- ✓ Profits: difference between F (agreed forward price, it was \$80 in the example above) and the spot market price S_T on the day of maturity T (which would have been attained otherwise):

$$\pi_{\text{short}} = F - S_T$$

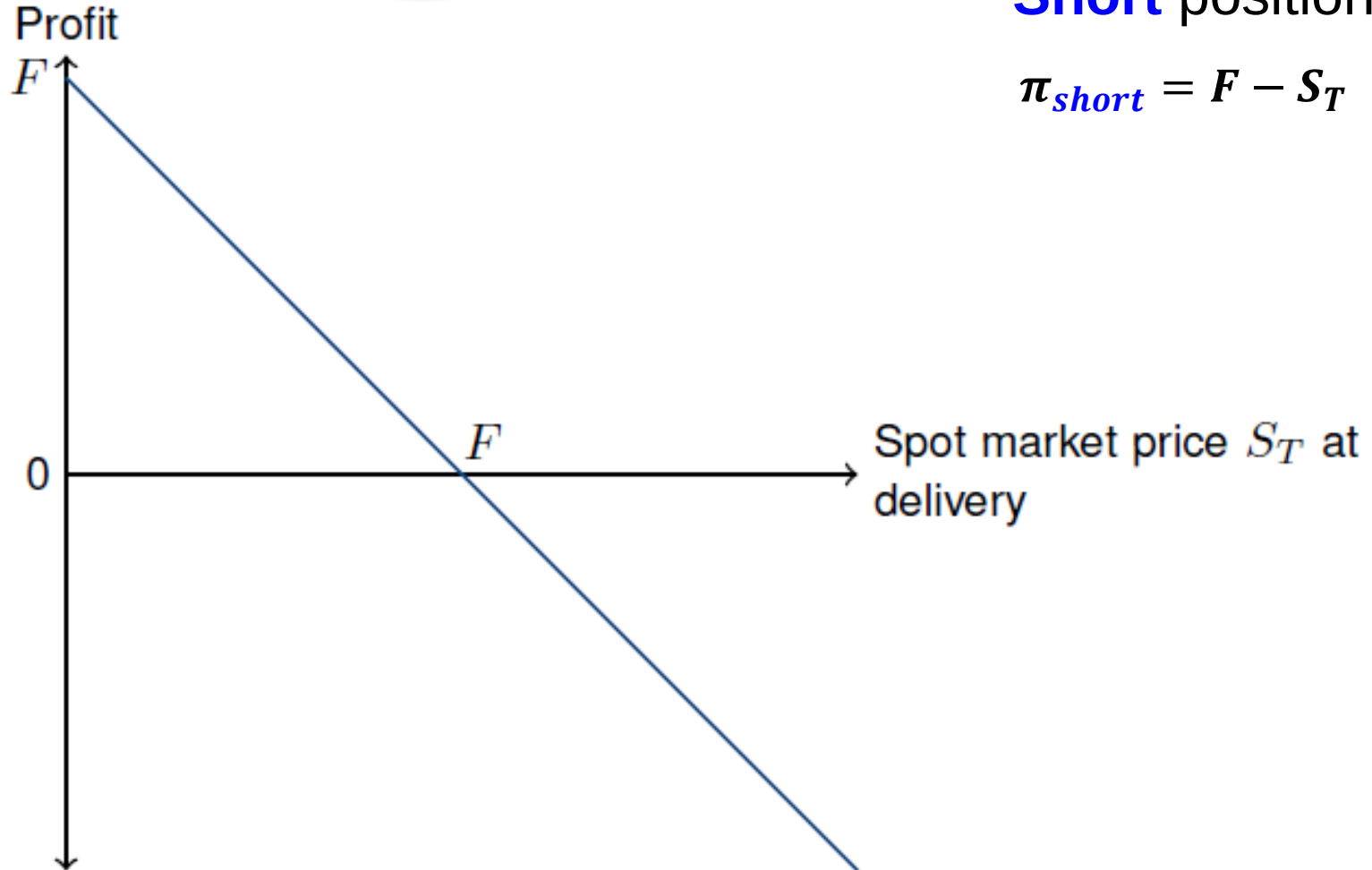
- ✓ **Selling** a forward contract *opens* a **short** position ...
 - ... which is the more valuable the lower the spot price of the asset will be in the future (next slide)

Profitability of a Forward Contract

Profits of the party selling a forward contract:

Short position

$$\pi_{short} = F - S_T$$



Profitability of a Forward Contract

- ✓ Profits of the party buying a forward contract:
 - in our example, this was the Utility company
- ✓ Profits: difference between the value of the item at the delivery time (spot price S_T), the party would have attained without the contract, and F :

$$\pi_{\text{long}} = S_T - F$$

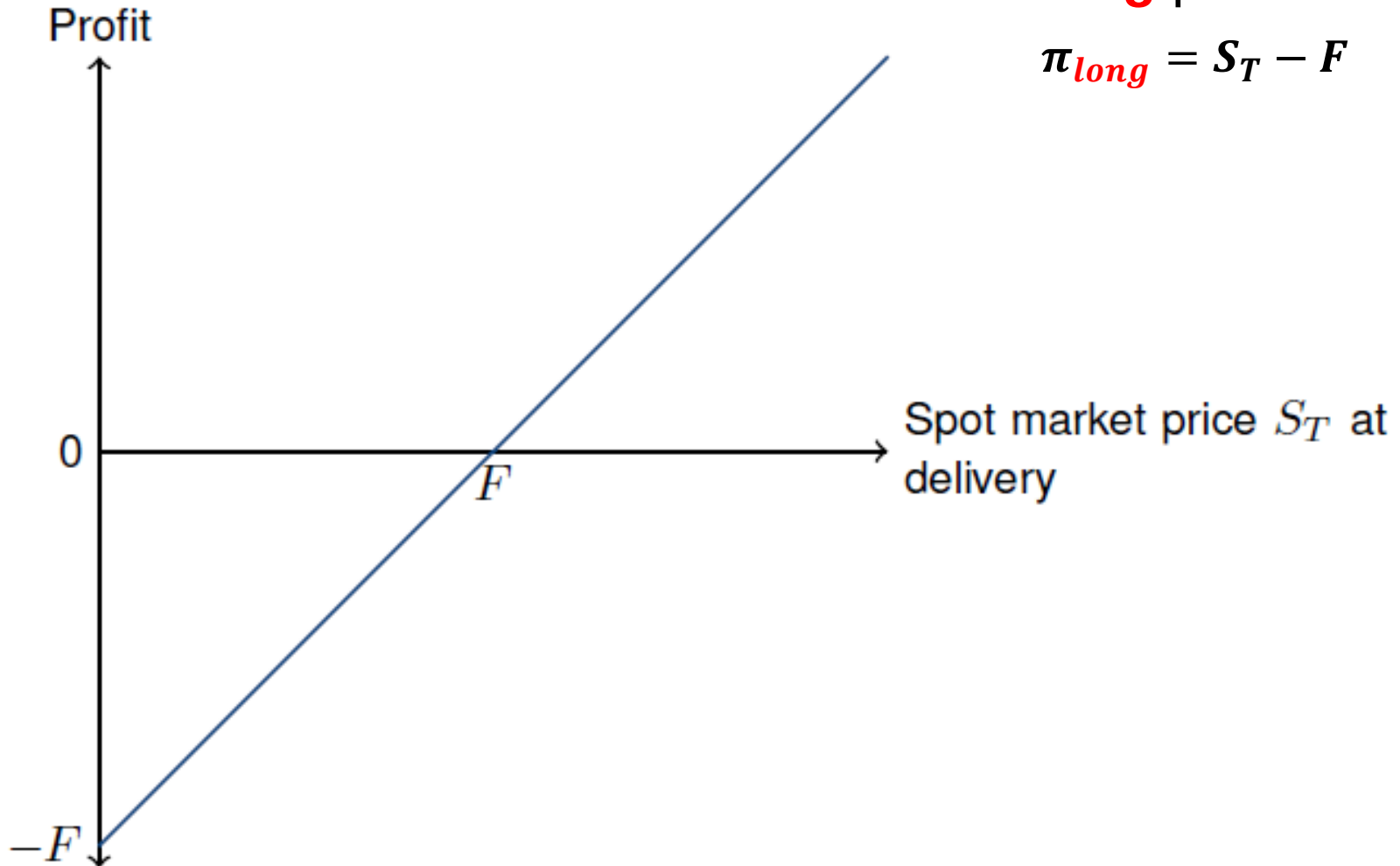
- ✓ **Buying** a forward contract opens a **long** position ...
 - which is the more valuable the higher the spot price of the asset will be in the future (next slide)

Profitability of a Forward Contract

Profits of the party buying a forward contract:

Long position

$$\pi_{long} = S_T - F$$



Forward Contract (Financial)

- ✓ There are of course forwards not only on real assets, but also on *financial assets* (4 ex. debt contracts)
- ✓ Example of an **Interest-Rate forward contract**
 - A bank agrees to deliver \$5 million in face value of 6% (coupon rate) Treasury bonds maturing in 2032
⇒ say: bank **sells** a bond forward
 - An insurance company agrees to pay \$5 million for the bonds **at the same interest rate as today**: 6%
⇒ say: insurance **buys** a bond forward
 - The bank and the insurance agree to complete the transaction **one year from today** (*delayed delivery*)

Forward Contract (Financial)

Q: who has taken a long position on the forward contract?
And the short?

Q: What risk does the bank want to hedge?

Q: What risk does the insurance want to hedge?

Forward Contract (Financial)

Q: who has taken a long position on the forward contract?
And the short?

- the bank (seller) has taken a **short** position
- the insurance company (buyer) has taken a **long** position

Q: What risk does the bank want to hedge?

Q: What risk does the insurance want to hedge?

Forward Contract (Financial)

Q: who has taken a long position on the forward contract?
And the short?

- the bank (seller) has taken a **short** position
- the insurance company (buyer) has taken a **long** position

Q: What risk does the bank want to hedge?

- that bond price may fall (int. rate may go up). So, they **sell** a forward at today's rate 6%

Q: What risk does the insurance want to hedge?

Forward Contract (Financial)

Q: who has taken a long position on the forward contract?
And the short?

- the bank (seller) has taken a **short** position
- the insurance company (buyer) has taken a **long** position

Q: What risk does the bank want to hedge?

- that bond price may fall (int. rate may go up). So, they **sell** a forward at today's rate 6%

Q: What risk does the insurance want to hedge?

- the insurance wants to invest in T-bonds, but they worry that price may rise (int. rate may go down). So, they **buy** a forward

Forward Contracts

Pros

1. Not standardized, so **flexible** (customized)

Cons

Since Forwards are traded on OTC markets

1. **lack of liquidity/depth:**

- hard to find a counter-party for the same contract: who would want exactly \$5M of 6% 2032 T-bonds?

2. **counterparty risk:** subject to default risk of the counterparty

3. **non-transferability:** it may be difficult to transfer to other parties (in the secondary markets) in case you need to avoid delivery for some reason

FUTURES were born to address these shortcomings!

Futures

From problems with forwards to solutions with Futures

1. lack of liquidity/depth

- design a *standardized* contract that trades on an organized exchange (rather than OTC)

2. counterparty risk

- introduce a mechanism to automatically protect against counterparty default

3. non-transferability

- introduce multilateral netting of positions in a “*clearinghouse*” (we will see how)

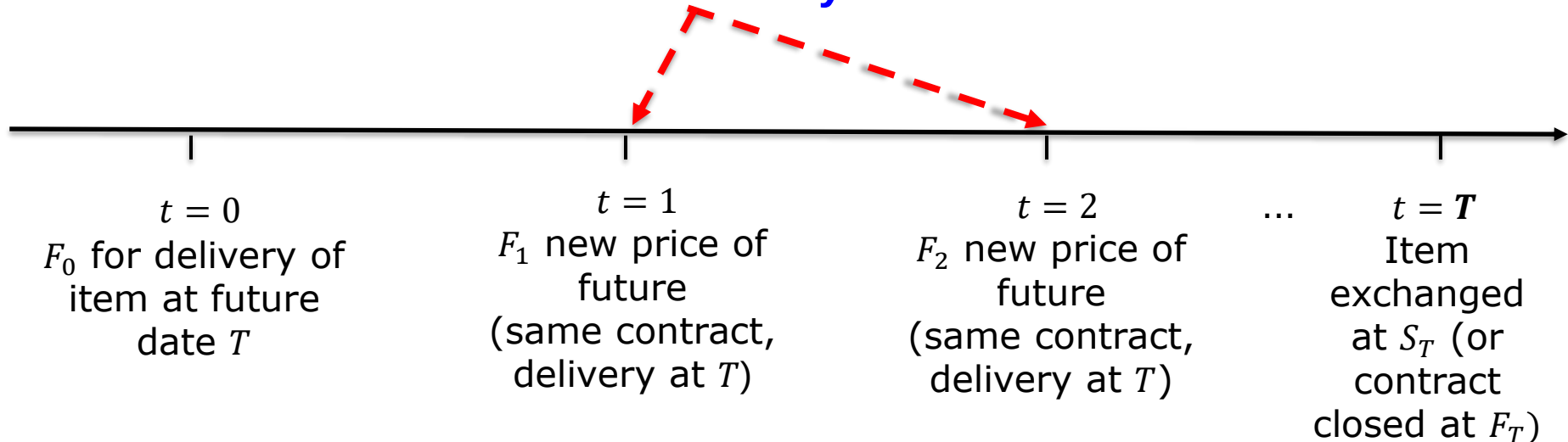
Futures

✓ Futures contracts:

- similar to forwards (contracts that specify an agreement to exchange an asset a future date at a price specified today)

✓ Differently from forwards, they are:

1. traded on **Organized Exchange** (4ex: *Chicago Mercantile Exchange*)
2. **standardized** in *amounts*, in *quality* and in *date* of delivery
3. “trading on an exchange” means that you can get a **price** for the future contract even **before delivery**:



Futures: Standardization

- ✓ Standardization overcomes the problem of **market liquidity** (*i.e.* find a counterparty) that forwards have
- ✓ Futures are standardized in much detail (such that it can be verified by court). Mainly:
 - **Contract size.** 4ex.: \$100,000 face value of T-bonds
 - **Delivery dates.** 4ex.: last business days of each quarter: March, June, September, December in the Chicago Board of Trade.
 - **Quality:** very specific. 4ex.: orange juice futures need to use this definition for orange juice:

“US Grade A, with Brix value of not less than 57 degrees, having a Brix value to acid ratio of not less than 13 to 1 nor more than 19 to 1, with factors of color and flavor each scoring 37 points or higher and 19 for defects, with a minimum score 94.”

Futures: Delivery

- ✓ The delivery of items in futures market almost never actually happens
Why?
- ✓ Because most traders
 - choose to **close their position** before the delivery period
 - trader “closes a position” by entering another trade of opposite type than the original one; i.e.: an offsetting position with the **exchange** directly
- ✓ the exchange (=clearing house ... an intermediary)
 - recognizes you both receive and deliver the same asset on the same date (so your net position is zero)
 - and so, it closes out your position: cancels both contracts
- ✓ this closing avoids the cost of delivery (transaction cost)

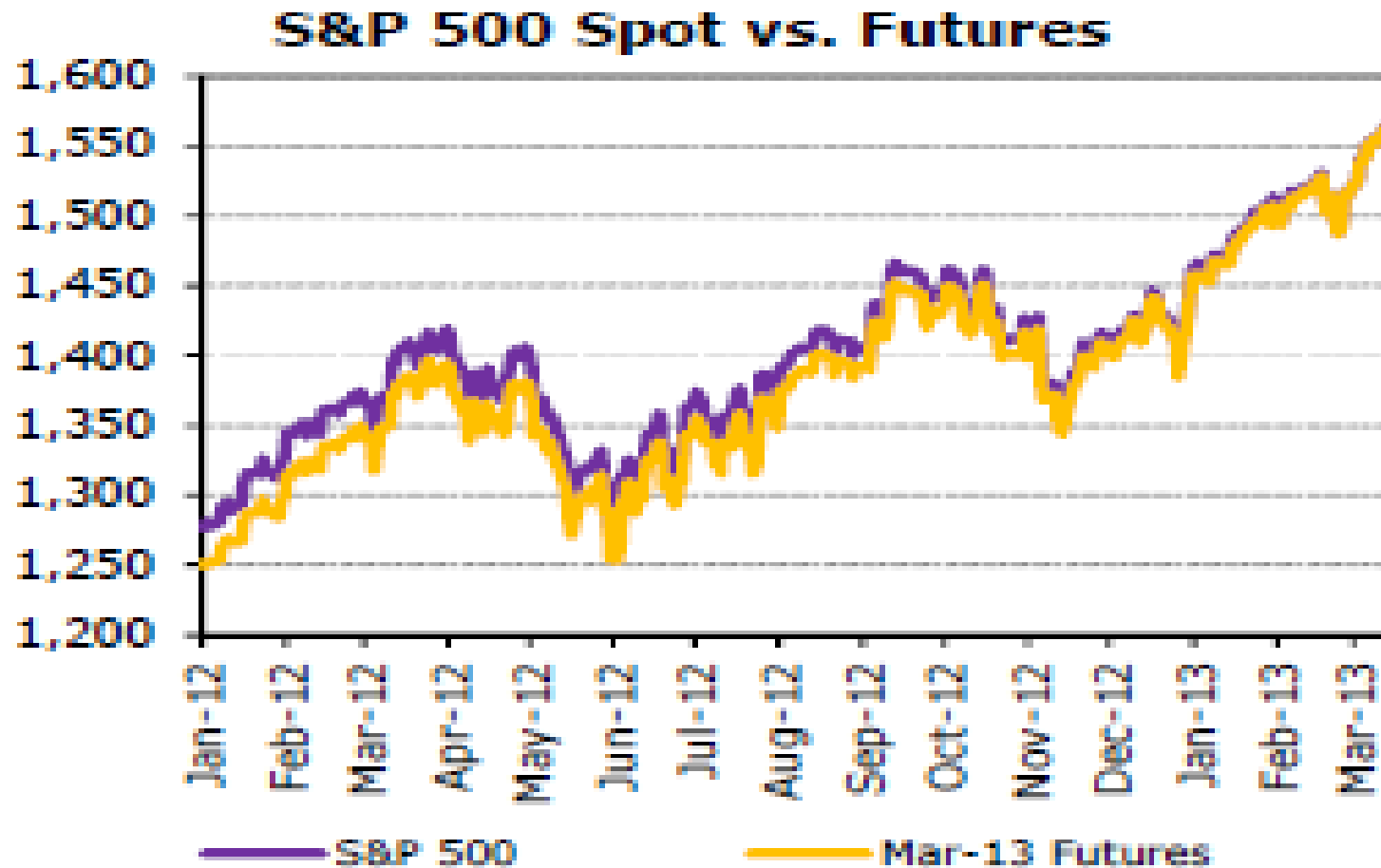
Example

- ✓ If you bought a future (=you are **long**)
... you can enter an additional position **short**
 - (sell the same future) with the exchange (=clearing house)

Futures: Prices

- ✓ An important fact: the **future price will converge to the spot price** at the **delivery date** ($F_T = S_T$)
- ✓ Why? Because otherwise there is an **arbitrage opportunity**:
 - if $F_T < S_T$ (price of a future for delivery today is **lower** than spot price), we can *buy* a future today, get the good delivered and then *sell* it immediately for profit at the spot price ($= S_T - F_T$): demand of futures $\uparrow \Rightarrow$ price of future (F_T) $\uparrow$
 - if $F_T > S_T$ (future price for delivery today is **above** spot price), buy at spot and *sell* futures ($= F_T - S_T$): supply of futures $\uparrow \Rightarrow$ (F_T) price of future $\downarrow$
- ✓ “Smart” traders will be quick to take advantage of any arbitrage opportunity and thus and eliminate them (**arbitrage**)

Basis Convergence (Basis = $F - S$)



Why Future prices move? Movements are largely driven by changes in expectations about the future demand and supply of the asset

Futures: Payoffs before/at expiration

- ✓ **Payoff** to the **long** side of a future contract closed at date t :

$$F_t - F_0$$

- where F_0 is the price initially paid on the future ...
- ... and F_t is the price of the future when closing (selling) before delivery

- ✓ **Payoff** to the **short** side of a future contract closed at date t :

$$F_0 - F_t$$

- where F_0 is the price initially received on the future ...
- ... and F_t is the price of the future when closing (buying) before delivery

- ✓ If the contract is kept until delivery ($t = T$): $F_t = F_T = S_T$

- but recall that $F_t \neq F_0$

Example: Profits going Long

- ✓ Assume that you took a long position on a futures contract when the price was **\$50**
- ✓ Today the futures price is **\$40**

Q1: What is your profit/loss if you close your position today?

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is even lower (say **\$30**)?

Example: Profits going **Long**

- ✓ Assume that you took a long position on a futures contract when the price was **\$50**
- ✓ Today the futures price is **\$40**

Q1: What is your profit/loss if you close your position today?

A1: If you close today (t) you need to go **short** (sell) at a lower price ($F_t = \$40$). Book a loss of $F_t - F_0 = \$40 - \$50 = -\$10$

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is *even lower* (say **\$30**)?

Example: Profits going **Long**

- ✓ Assume that you took a long position on a futures contract when the price was **\$50**
- ✓ Today the futures price is **\$40**

Q1: What is your profit/loss if you close your position today?

A1: If you close today (t) you need to go **short** (sell) at a lower price ($F_t = \$40$). Book a *loss* of $F_t - F_0 = \$40 - \$50 = -\$10$

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is *even lower* (say **\$30**)?

A2: If wait until delivery, you buy at $F_0 = \$50$ something that sells for $S_T = \$30$. You *lose*: $S_T - F_0 = \$30 - \$50 = -\$20$

Example: Profits going **Short**

- ✓ Everything the opposite if we are **short** at $t = 0$ on same future
- ✓ Assume that you took a **short** position on a futures contract when the price was **\$50**
- ✓ Today the futures price is **\$40**

Q1: What is your profit/loss if you close your position today?

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is *even lower* (say **\$30**)?

Example: Profits going Short

- ✓ Everything the opposite if we are **short** at $t = 0$ on same future
- ✓ Assume that you took a **short** position on a futures contract when the price was **\$50**
- ✓ Today the futures price is **\$40**

Q1: What is your profit/loss if you close your position today?

A1: If you close today (t) you need to go **long** (buy) at a lower price ($F_t = \$40$). Book a *profit*: $F_0 - F_t = \$50 - \$40 = +\$10$

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is *even lower* (say **\$30**)?

Example: Profits going Short

- ✓ Everything the opposite if we are short at $t = 0$ on same future
- ✓ Assume that you took a short position on a futures contract when the price was \$50
- ✓ Today the futures price is \$40

Q1: What is your profit/loss if you close your position today?

A1: If you close today (t) you need to go long (buy) at a lower price ($F_t = \$40$). Book a *profit*: $F_0 - F_t = \$50 - \$40 = +\$10$

Q2: What if you wait until the delivery/settlement date, when the price of the underlying asset is *even lower* (say \$30)?

A2: If wait until delivery, you sell at $F_0 = \$50$ something that you may buy for $S_T = \$30$. You *profit*: $F_0 - S_T = \$50 - \$30 = +\$20$

Futures: Exchanges & Default Risk

- ✓ The exchange deals with the other big issue of a forward contract: **default risk** of the counterparty
- ✓ How?
- ✓ The exchange acts as a *clearinghouse* (**CCP, Central CounterParty**) for both traders:
 - the clearinghouse **guarantees** all futures will be executed even if one counterparty defaults
- ✓ Where does it get the money?
 - from the buyers and sellers **margin accounts**, whose amount is tied to the value of the future contract

Futures: Margin Account

✓ The **margin account** (m.a.)

- similar to a bank account to which the clearinghouse of the futures exchange can make automatic debits/credits, depending on how trader's futures contract evolves...
- the value of the margin account changes as the value of the future contract changes
- the account has
 - an "*initial margin*"
 - and a minimum "*maintenance requirement*"
- if your margin account has *insufficient funds* (it runs below the maintenance requirement), the exchange will give you a margin call asking you to top up with more money (or they will automatically close your position)
 - if the balance of your margin account *exceeds* the "initial margin requirement", you may withdraw those excess funds at anytime

Futures: Marking-to-market

- ✓ Key point: *m.a.* is adjusted *daily* (**daily settlement**):
 - at each closing date, market prices determine the value in each margin account: **marking-to-market**
 - this eliminates **default risk**. Let's see how it works ...
- ✓ At the end of every day, the exchange looks at how futures prices have moved
 - it **credits** the margin account of the party whose contract has become more valuable ...
 - .. and it **debits** the counterparty's account by exactly the same amount (because contract has become less_valuable)
- ✓ The **amount** debited/credited corresponds
 - to the change in futures prices between **today** and **yesterday**: F_t Vs F_{t-1}
 - ... considering the quantity of the good to be delivered

Margin calls can cause problems

Uniper and Wien Energie seek state support as energy crisis deepens

German utility asks for €4bn while Austria's largest supplier admits to difficulty financing operations



“The rapid run-up in prices had further increased the margin payments — or collateral — Uniper was required to make to exchanges when it trades gas and power” ([FT, August 29, 2022](#))

Exercise 1: Mark-to-Market (1/4)

- ✓ Suppose a trader has taken a **long** position in oil futures at **\$75** per barrel
 - the contract expires in 3 days, and it consists of 5,000 barrels
 - as *initial margin* the trader needs to deposit 4% of value of future contract
 - value of future contract = $Q \times F_0$ (=contract size x future price)
- ✓ The futures price in the following three days are:
 - 1st day: $F_1 = \$80$
 - 2nd day: $F_2 = \$74$
 - 3rd day: $F_3 = \$77$

Q: Calculate the trader's profits and losses in her m.a.

Exercise 1: Mark-to-Market (2/4)

- ✓ The trader is **long** in oil futures
 - initial margin = $Q \times F_0 \times im(\%) = 5,000 \times \$75 \times 0.04 = \$15,000$
- ✓ If the price **increases** ($\Delta F > 0$), trader in **long** *profits*
 - clearing house credits the additional profit due to price increase ($Q \times \Delta F$) on the margin account of trader in long (debits the same amount to trader in short)
- ✓ If the price **decreases** ($\Delta F < 0$), trader in **long** has *loss*
 - clearing house debits $Q \times \Delta F$ on long trader's margin account

Exercise 1: Mark-to-Market (2/4)

- ✓ The trader is **long** in oil futures
 - initial margin = $Q \times F_0 \times im(\%) = 5,000 \times \$75 \times 0.04 = \$15,000$
- ✓ If the price **increases** ($\Delta F > 0$), trader in **long profits**
 - clearing house credits the additional profit due to price increase ($Q \times \Delta F$) on the **m.a.** of trader in long (debits the same amount to trader in short)
- ✓ If the price **decreases** ($\Delta F < 0$), trader in **long has loss**
 - clearing house debits $Q \times \Delta F$ on long trader's **m.a.**

Day	F_t	Future Price ($\Delta F = F_t - F_{t-1}$)	Cash flow to m.a.	m.a. balance
0	\$75	--	--	\$15k
1	\$80	\$5	+\$25k = (5k × (+\$5))	\$40k (= \$15k + \$25k)
2	\$74	-\$6	-\$30k = (5k × (-\$6))	\$10k (= \$40k - \$30k)
3	\$77 (= S_T)	\$3	+\$15k = (5k × (+\$3))	\$25k (= \$10k + \$15k)
Total		+\$2	+\$10k = (5k × (\$77 - \$75))	\$10k (= \$25k - \$15k)

Exercise 1: Mark-to-Market (3/4)

- ✓ Since the position was not closed before maturity, delivery takes place at the **spot price of 77\$** (not at \$75):
 - Note delivery at F_T , not at F_0 !
- ✓ This is also the price of a future bought at the delivery date (convergence of future with spot price)
- ✓ But wait! She **bought** oil futures for \$75/bbl, as agreed 3 days before, and not \$77

Q: where are those 2\$/bbl?

A: in the **margin account!**

Exercise 1: Mark-to-Market (4/4)

- ✓ In fact, on the **futures position** the **long** trader gained:

$$\pi_{long} = (S_T - F_0) \times Q = (\$77 - \$75) \times 5,000 = +\$10,000$$

- ✓ This is **also** the **same** as the **total cash flows in the margin accounts** $(\$25k - \$30k + \$15k) = +\$10k$

- ✓ So, the trader pays \$77 for barrels, but also gets +\$10,000 in the margin account which is **like paying \$75** for each barrel:

$$-\$77 + \frac{\$10,000}{5,000(bbl)} = -\$77 + \$2 = -\$75$$

$$[\text{or can see it as } \frac{\$10,000}{5,000(bbl)} = \$2 = \$77 - \$75]$$

Exercise 2: Mark-to-Market (1/4)

- ✓ Suppose a trader has taken a **short** position in silver futures at **\$1,220** per ounce
 - the contract expires in 2 days, and it consists of 100 ounces
 - the trader has to
 - deposit in the m.a. 4% of value of future contract as *initial margin*
 - keep in the m.a. 3% of value of future contract as *maintenance margin*
 - value con future contract = $Q \times F_0$ (=contract size x future price)

- ✓ The futures price in the following two days are:
 - 1st day: $F_1 = \$1,200$
 - 2nd day: $F_2 = \$1,205$

Q: Calculate the trader's profits and losses in her m.a.. Does trader receive margin call to top up funds?

Exercise 2: Mark-to-Market (2/4)

- ✓ The trader is **short** in silver futures
 - initial margin = $Q \times F_0 \times im(\%) = 100 \times \$1,220 \times 0.04 = \$4,880$
- ✓ If the price **increases**, trader in **short** has loss
 - clearing house debits of $Q \times \Delta F$ on short trader's margin account
 - if sum is lower than maintenance margin [**maintenance margin** = $Q \times F_0 \times mm(\%) = 100 \times \$1,220 \times 0.03 = \$3,600$] clearing house makes a margin call to top up funds in the m.a.
- ✓ If the price **decreases**, trader in **short** profits
 - clearing house credits the additional profit ($Q \times \Delta F$) on the margin account of trader in short (debits the same amount to trader in long)

Exercise 2: Mark-to-Market (2/4)

✓ The trader is **short** in silver futures

○ **initial** margin = $Q \times F_0 \times im(\%) = 100 \times \$1,220 \times 0.04 = \underline{\$4,880}$

○ **maintenance** margin = $Q \times F_0 \times mm(\%) = 100 \times \$1,220 \times 0.03 = \underline{\$3,600}$

Day	F_t	Future Price ($\Delta F = F_{t-1} - F_t$)	Cash flow to m.a.	m.a. balance
0	\$1,220	--	--	\$4,880
1 st	\$1,200	+\$20	+\$2,000 (= 100 x \$20)	\$6,880 (= \$4,880 + \$2,000)
2 nd	\$1,205 (= S_T)	-\$5	-\$500 (= 100 x (-\$5))	\$6,380 (= \$6,880 - 500)
Total		+\$15	+\$1,500 (= 100 x (\$1,220 - \$1,205))	+\$1,500 (= \$6,380 - \$4,880)

✓ Trader does not receive margin call

○ indeed m.a. never falls below \$3,600

Exercise 2: Mark-to-Market (3/4)

- ✓ Since the position was not closed before maturity, delivery takes place at the **spot price** (S_T) **1,205\$** (not at \$1,220):
 - Note delivery at F_T , not at F_0 !
- ✓ This is also the price of a future sold at the delivery date (again, convergence of future with spot price)
- ✓ Even if she **sold** silver futures for \$1,220/ounce, as agreed 2 days before, trade is at \$1,205

Q: where are those 15\$/ounce?

A: in the **margin account**!

Exercise 2: Mark-to-Market (4/4)

- ✓ In fact, on the **futures** position the **short** trader **gained**:

$$\pi_{\text{short}} = (F_0 - S_T) \times Q = (\$1,220 - \$1,205) \times 100 = +\$1,500$$

- ✓ This is also the same as the total cash flows in the margin accounts: $\$2,000 - \$5,000 = \$1,500$

- ✓ So, the trader gets \$1,205 for ounce, but also gets +\$1,500 in the margin account which is **like paying \$1,220** for each ounce:

$$\$1,205 + \frac{\$1,500}{100(\text{ounces})} = \$1,205 + \$15 = \$1,220$$

$$[\text{or can see it as } \frac{\$1,500}{100(\text{ounces})} = \$15 = \$1,220 - \$1,205]$$

The Elimination of Counterparty Credit Risk

Credit risk is no longer an issue. Why?

- ✓ Suppose a counterparty defaults, no big problem:
 - its position will be closed immediately
 - the entire value which the contract had at the moment of the default has already been paid by the now-defaulting counterparty via the margin account system
 - so, you could simply look for another contract partner for your futures position ...
 - ... simple because the exchange will do that, even automatically, for you: you suffered **no loss at all!**

Profits of short/long investors in futures

✓ Before going on, let's recall each party's profit function

✓ Long investor

$$\pi_{long,t} = Q \cdot [(F_1 - F_0) + (F_2 - F_1) + \dots + (F_t - F_{t-1})]$$

$$\pi_{long,t} = Q \cdot (F_t - F_0)$$

○ if wait delivery, since $F_T \rightarrow S_T$, the total profit/loss to long investor is:

$$\pi_{long,T} = Q \cdot (S_T - F_0)$$

✓ Short investor

$$\pi_{short,t} = Q \cdot [(F_0 - F_1) + (F_1 - F_2) + \dots + (F_{t-1} - F_t)]$$

$$\pi_{short,t} = Q \cdot (F_0 - F_t)$$

○ if wait delivery, since $F_T \rightarrow S_T$, the total profit/loss to short investor is:

$$\pi_{short,T} = Q \cdot (F_0 - S_T)$$

Stock Index Future

- ✓ In a **stock index future**, the underlying asset is a stock market index (4 ex: S&P500)
- ✓ Useful to hedge **stock market risk**
- ✓ What item is delivered in this case? Just **cash!**
 - not a security as in other financial futures
- ✓ How much? The cash amount is equal to the stock market index at the delivery date $\times$ ***multiplier***

$$\pi_{\text{long}, T} = \text{multiplier} \times (S_T - F_0)$$

$$\pi_{\text{short}, T} = \text{multiplier} \times (F_0 - S_T)$$

Example: S&P500 futures

- ✓ For the S&P500 futures the multiplier is 250
- ✓ If you sell one contract at a price $(S\&P500_T)$ of \$1,500, you agree to deliver in the future $250 \times (S\&P500_T)$ at the delivery date:
 - value of futures contract: \$375,000 ($= 250 \times \$1,500$)
- ✓ Suppose the spot price at $t = T$ is: $S\&P500_T = \$1,000$:
 - $$\pi_{short,T} = 250 \times (\$1,500 - \$1,000)$$
$$\pi_{short,T} = 250 \times \$500 = \$125,000$$
 - you sold at \$375k but only need to deliver \$250k ($= \$375k - \$125k$)
 - technically, only \$125k will be exchanged at delivery

Stock Index Future

Why would you use a stock index future?

- ✓ Suppose you are long in a portfolio of stocks of \$100M that moves one-to-one with the S&P500
- ✓ you want to hedge a downturn: go **short** on S&P500
- ✓ hedge for one year
- ✓ the S&P500 futures is currently selling at \$1,000

Q: How many future contracts do you need?

$$\# \text{ contracts} = \frac{\text{Value of Portfolio}}{\text{Value of S\&P futures}} = \frac{\text{Value of Portfolio}}{(S\&P 500_T \times \text{multiplier})}$$

$$\# \text{ contracts} = \frac{\$100,000,000}{(\$1,000 \times 250)} = \frac{\$100,000,000}{\$250,000} = 400$$

Stock Index Futures

- ✓ Suppose after the year, the S&P 500 is at \$900
- ✓ You had a **short** position in the future, so you gained:
 - $\pi_{\text{short}} = \Delta F \times (\# \text{contracts}) \times \text{multiplier}$
 - $\pi_{\text{short}} = (\$1,000 - \$900) \times 400 \times 250 = +\$10M!$
- ✓ this will also be the total inflow in the m.a.
 - since this is cash settled, you never actually sell a portfolio of stocks at \$900, but only get credited the difference (+\$10M) in your m.a.
- ✓ But your portfolio of \$100M is now worth: \$90M
- ✓ So, your **total net position** (*portfolio+future profit*) =
\$90M + \$10M = **\$100M**

Stock Index Futures

- ✓ Suppose instead, after the year the S&P 500 is at \$1,100
- ✓ You had a *short* position, so you lost:
 - $\pi_{\text{short}} = \Delta F \cdot (\#contracts) \cdot multiplier$
 - $\pi_{\text{short}} = (\$1,000 - \$1,100) \cdot 400 \cdot 250 = -\$10M!$
- ✓ this will also be the total outflow from the m.a.
 - again, it is cash settled; you never sell a portfolio of stocks at \$1,100, but only get debited the difference (-\$10M) in your m.a.
- ✓ But your portfolio of \$100M is now worth more: \$110M
- ✓ So, your **total net position** (*portfolio+future profit*)=
\$110M - \$10M = **\$100M**

Stock Index Futures

Either way, net position is \$100 million

- ✓ So, you can gain/lose on futures but opposite for portfolio: that's what the hedge is for!
- ✓ The portfolio is protected from **downside** risk
- ✓ However, to accomplish this, the manager has also eliminated any **upside** potential
- ✓ **Important lesson about hedging:** great to have it, but a hedge gone bad is very dangerous

Summary: Forwards vs Futures

	Forward Contract	Futures Contract
<i>Traded</i>	over-the-counter	on exchange
<i>Standardization</i>	none	standardized
<i>Customizability</i>	arbitrary	none
<i>Delivery</i>	usually takes place	rarely held until delivery
<i>Settlement</i>	at end of contract	daily (marked-to-market)
<i>Credit risk</i>	some	virtually none
<i>Liquidity</i>	low	very high